

Theory note S1 — Frequency hopping under partial-band jamming: processing gain, the worst-case band fraction, and the jamming margin

Chang Young Ham

September 2026 (accompanies `src/s1_fh_partial_band_jamming.py`)

Purpose. S1 adds a hopping transmitter and an intelligent jammer to the S0 link. This note derives (i) why hopping buys a processing gain, (ii) the closed-form BER used to verify the simulator, (iii) the jammer's optimal band fraction ρ^* and the resulting inverse-linear BER law, and (iv) the jamming margin numbers reported in `results/s1_link_budget.txt`. Mathematics used: total probability, one-variable optimisation, a first-order asymptotic of the Gaussian tail.

1 Processing gain

The data occupies bandwidth W_d ; the transmitter hops among N_{ch} channels spread over $W_{ss} \approx N_{\text{ch}}W_d$. A jammer with total power J that wants to cover the whole hopping band must spread J over W_{ss} , so its density is $N_J = J/W_{ss}$. The transmitter's signal at any instant occupies only W_d , where the jammer delivers only $N_JW_d = JW_d/W_{ss} = J/N_{\text{ch}}$ of its power. The jammer effectively loses a factor

$$G_p = \frac{W_{ss}}{W_d} \approx N_{\text{ch}}, \quad G_p [\text{dB}] = 10 \log_{10} N_{\text{ch}}.$$

With $N_{\text{ch}} = 100$: $G_p = 20 \text{ dB}$.

Relating E_b/N_J to J/S . With signal power S and bit rate R_b , $E_b = S/R_b$. Taking $W_d \approx R_b$,

$$\frac{E_b}{N_J} = \frac{S/R_b}{J/W_{ss}} = \frac{S}{J} \cdot \frac{W_{ss}}{R_b} \approx \frac{S}{J} G_p \quad \implies \quad \boxed{\frac{E_b}{N_J} [\text{dB}] = G_p [\text{dB}] - \frac{J}{S} [\text{dB}]}.$$

This is the line `ebnj_lin = db_to_linear(gp_db - js_db)` in the code and the reason the figure's x -axis is J/S while the physics runs on E_b/N_J .

2 Partial-band jamming: the mixture formula

An intelligent jammer does not spread over the whole band. It concentrates J on a fraction $\rho \in (0, 1]$ of the channels, raising its density there to N_J/ρ and leaving the rest clean. The transmitter does not know which channels are jammed (pseudo-random pattern), so each hop lands in the jammed region independently with probability ρ . By the law of total probability,

$$P_b(\rho) = (1 - \rho) \text{Q}\left(\sqrt{\frac{2E_b}{N_0}}\right) + \rho \text{Q}\left(\sqrt{\frac{2E_b}{N_0 + N_J/\rho}}\right). \quad (1)$$

The first term is the thermal-noise-only BER of S0 (clean hop); the second is an S0 link whose noise density is N_0 plus the concentrated jammer density. This is `ber_partial_band_theory()` and it is what the Monte Carlo (random jammed flag per hop, then the S0 machinery) is checked against.

Limit check ($\rho = 1$). Eq. (1) reduces to $Q(\sqrt{2E_b/(N_0 + N_J)})$: plain AWGN with the two densities added. VERIFICATION 2 asserts exactly this (agreement to 10^{-12}).

3 The jammer's optimal ρ and the inverse-linear law

From the jammer's point of view ρ is a control knob and it wants to maximise (1). When the jammer dominates the thermal noise ($N_J/\rho \gg N_0$), drop N_0 :

$$P_b(\rho) \approx \rho Q\left(\sqrt{\frac{2\rho E_b}{N_J}}\right).$$

Substitute $x = 2\rho E_b/N_J$ (so $\rho = x/(2E_b/N_J)$):

$$P_b \approx \frac{x Q(\sqrt{x})}{2 E_b/N_J} = \frac{g(x)}{2 E_b/N_J}, \quad g(x) := x Q(\sqrt{x}).$$

The dependence on E_b/N_J has been pulled out; the jammer just maximises $g(x)$ over $x > 0$. Setting $g'(x) = 0$, with $Q'(u) = -\varphi(u)$ and $\varphi(u) = e^{-u^2/2}/\sqrt{2\pi}$,

$$g'(x) = Q(\sqrt{x}) - x \varphi(\sqrt{x}) \frac{1}{2\sqrt{x}} = Q(\sqrt{x}) - \frac{\sqrt{x}}{2} \varphi(\sqrt{x}) = 0.$$

This transcendental equation has one positive root, found numerically:

$$x^* \approx 1.418, \quad g(x^*) \approx 0.1657.$$

Stop here and check: evaluate both sides of $Q(\sqrt{x}) = \frac{\sqrt{x}}{2} \varphi(\sqrt{x})$ at $x = 1.418$; they agree to four decimals (0.1169).

Two consequences follow immediately.

$$\rho^* = \frac{x^*}{2 E_b/N_J} \approx \frac{0.709}{E_b/N_J}, \quad (2)$$

$$P_b^{\text{worst}} = \frac{g(x^*)}{2 E_b/N_J} \approx \frac{0.083}{E_b/N_J}. \quad (3)$$

Eq. (3) is the central fact of S1: *against an optimal partial-band jammer the BER decays only as $1/(E_b/N_J)$* , not exponentially as in S0. Doubling the signal power halves the error rate instead of squaring it away. (Eq. (2) is only meaningful while $\rho^* \leq 1$, i.e. $E_b/N_J \geq x^*/2 \approx -1.5$ dB; below that the jammer simply uses the whole band.)

The number in the figure. At $J/S = 12$ dB with $G_p = 20$ dB, $E_b/N_J = 8$ dB = 6.31. Eq. (2) gives $\rho^* = 0.709/6.31 = 0.112$ with thermal noise ignored. Keeping the thermal term N_0 in (1) shifts the optimum slightly upward; a numerical search of the exact formula gives $\rho^* = 0.135$, which is the 0.14 reported by both the closed-form grid and the Monte Carlo (VERIFICATION 3). The corresponding worst-case BER is 1.42×10^{-2} versus 2.70×10^{-3} for a full-band jammer of the same power: the factor of five in the figure.

4 Jamming margin

A link is declared *up* if $P_b \leq \text{BER}_{\text{out}} = 10^{-3}$ (assumed criterion; the S2 note discusses its consequences). Invert the BER formula to find the E_b/N_J at which the criterion is just met, $(E_b/N_J)_{\text{req}}$, then allow an implementation loss L_{sys} . The jamming margin is the J/S the link can absorb:

$$M_J [\text{dB}] = G_p - (E_b/N_J)_{\text{req}} - L_{\text{sys}}.$$

For $N_{\text{ch}} = 100$, $E_b/N_0 = 10 \text{ dB}$, $L_{\text{sys}} = 2 \text{ dB}$:

jammer strategy	$(E_b/N_J)_{\text{req}}$	M_J
full band, $\rho = 1$	9.65 dB	$20 - 9.65 - 2 = \mathbf{8.35 \text{ dB}}$
optimal ρ	18.95 dB	$20 - 18.95 - 2 = \mathbf{-0.95 \text{ dB}}$

The optimal jammer needs 9.3 dB *less* power to break the link. The *negative* margin in the second row means that at $\text{BER}_{\text{out}} = 10^{-3}$, 100 hop channels alone cannot protect a BPSK link against a jammer of equal received power — a statement that motivates every extension listed in S2.

5 Correspondence with the code

Mathematics	Code
$G_p = 10 \log_{10} N_{\text{ch}}$	<code>gp_db = linear_to_db(N_CH)</code>
$E_b/N_J = G_p - J/S$	<code>ebnj_lin = db_to_linear(gp_db - js_db)</code>
jammed hop w.p. ρ	<code>jammed = rng.random(...) < rho</code>
$\sigma^2 = (N_0 + N_J/\rho)/2$	<code>sigma_jammed = np.sqrt((n0 + nj / rho) / 2.0)</code>
Eq. (1)	<code>ber_partial_band_theory()</code>
ρ^* by grid search	<code>RHO_GRID[np.argmax(ber_rho_theory)]</code>
$(E_b/N_J)_{\text{req}}$	first grid point with worst-case $\text{BER} \leq 10^{-3}$

6 Self-check

1. Explain in one sentence why a jammer that covers the whole hopping band wastes $1 - 1/N_{\text{ch}}$ of its power.
2. Derive (1) from the law of total probability.
3. Starting from $P_b \approx \rho Q(\sqrt{2\rho E_b/N_J})$, show that the optimal ρ scales as $1/(E_b/N_J)$ and that the worst-case BER scales the same way. Why does this matter more than the exact constant 0.083?
4. Why is the required E_b/N_J for the optimal jammer (18.95 dB) so much larger than for the full-band jammer (9.65 dB)? Relate the gap to the $1/x$ versus e^{-x} decay laws.
5. What would change in the analysis if the jammer *knew* the hopping pattern (a follower jammer)? Which assumption breaks?